SKYWAVE™
AN ORBCOMM COMPANY



ST 8100
Installation Guide

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PREFACE

Purpose

This document is as an overview of the installation procedures for the ST 8100 (Standard and Advanced variant). The information in this guide refers to both variants unless noted otherwise.

Notation

Hardware components, labels, and figures in this document might not be exactly as shown and are subject to change without notice.

CAUTION: This safety symbol warns of possible hazards to personnel, equipment, or both. It includes hazards that will or can cause personal injury, property damage, or death if the hazard is not avoided.

Note: A note indicates information with no potential hazard. A note indicates points of interest or provides supplementary information about a feature or task.

Numbered lists indicate a series of steps required to complete a task or function.

Bulleted lists highlight information where order or sequence is not crucial.

Reference

The content of the following documents might be useful in conjunction with this guide. These documents are available from the downloads section of the partner support website or from the ORBCOMM Developer Toolkit (ODT), which is also available from the website.

Document names and numbers are subject to change, or be discontinued, without notice. Always check the partner support website for the most current version of these documents.

[T404]	LSF Developer Guide
[T405]	LSF API Reference Guide
[T416]	ST 8100 Hardware Guide

Battery Safety Warnings

CAUTION: DO NOT short circuit or expose the battery to temperatures above the maximum rated temperature.

CAUTION: Always follow local disposal guidelines to properly dispose of the Lithium-ion battery and the device.

CAUTION: Store in a cool, well-ventilated area. Elevated temperatures will result in shortened battery life.

CAUTION: DO NOT replace the battery. Changing the battery without ORBCOMM's permission could violate regulatory conformity.

CAUTION: DO NOT throw the internal battery or the device into fire.

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1 INSTALLATION

The following section contains recommended installation guidelines for the Solution Provider (SP). These recommendations should be incorporated into installation guidelines for end users.

CAUTION: The installer is responsible for following all safety guidelines during product installation. Refer to the Preface section for details.

IMPORTANT

READ ALL INSTRUCTIONS CAREFULLY BEFORE INSTALLING. FAILURE TO DO SO MAY CAUSE PERSONAL INJURY OR DAMAGE TO PRODUCT AND/OR PROPERTY.

• Review the product package and contents prior to beginning the installation. Take care when opening the packaging and removing items. If a return is needed, you will want to return the product in its original packaging if possible. • This instruction guide is provided as a GENERAL installation guide; some assets vary dimensionally and may require additional steps. • ORBCOMM has a policy of continuous development and improvements. Therefore, products, guides, and technical information are subject to change without prior notice. • The manufacturer and / or distributors do not accept responsibility for third-party charges, labor, and / or third-party replacement modifications that are not ORBCOMM approved. Some modifications may void the factory warranty. • ORBCOMM does not accept any responsibility for installations performed by installers / third parties not approved and / or authorized by ORBCOMM. Some installations may void the factory warranty. • Exercise due diligence when installing this product. ORBCOMM does not accept any responsibility for asset damage or personal injury resulting from the installation of this product. Careless installation and operation can result in serious injury or equipment damage. • All liability for installation and use rests with the owner / operator. • Always make sure you have a clean, dry, and well-lit work area. • Always ensure products are secure during disassembly and installation. • Always take steps to protect yourself when drilling, cutting, and grinding because this may create flying particles that can cause injury. • Thoroughly inspect the area to be drilled, on both sides of material, prior to modification, and relocate any objects that may become damaged. • Always route electrical cables carefully. Avoid moving parts, parts that may become hot and rough, or sharp edges. • Make sure to fully understand the product, its intended use, and operation prior to use.

CAUTION: While we provide mounting hardware to assist with installations, it is the responsibility of the installer to select the proper mounting hardware for the asset's surface material where a device or accessory will be mounted.

1.1 Wake Up the ST 8100

The ST 8100 ships in a non-transmitting mode. Prior to installation on an asset, you must wake up the device by exposing it to bright light.

Note: Expose the ST 8100 to bright light at the beginning of the day to ensure it has time to wake up and transmit critical information before you install it.

1. Remove the ST 8100 from the shipping packaging, and then wake up the device in one of two ways:
   - Bring the ST 8100 outside during daylight, with the solar panel facing the sky, for a minimum of 5 minutes.

     OR
   - Use another bright light source to wake up the device. Use a light meter or a mobile phone app to find a light source with a minimum 600 Lux light intensity, the higher the Lux value, the better. Position the ST 8100 solar panel directly facing the light source for at least one minute.

     CAUTION: When using a hot light source (for example, halogen or incandescent), ensure that the solar panel does not become hot to the touch.

   Note: The ST 8100 solar panel does not charge with typical indoor office lighting.

   Note: The green LED flashes on startup.

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1.1.1 Remove the PIN Label

1. Remove the BLE PIN label taped to the top of the ST 8100.

   CAUTION: DO NOT leave the label on the device. SKYWAVE recommends you save this PIN in a safe location for future use.



Note: (Optional) You can scan the QR Code on the label to access the BLE PIN (this is also the Solution Provider password) and device serial number required for Bluetooth pairing and access to restricted configuration functions. These credentials are unique to this device and can be changed after installation, if required

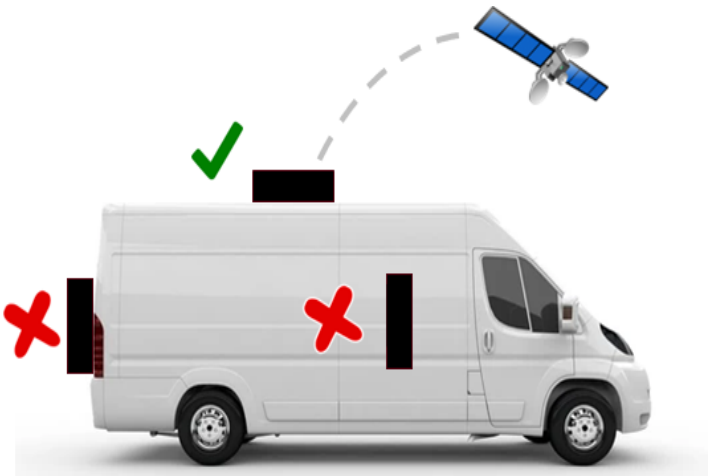
1.2 Prepare the Mounting Location

1. Determine the mounting location for the ST 8100 following the guidelines below:

   CAUTION: Mount the device at least 20 cm (8 in.) away from humans.

   - Mount the device where it has a clear view of the sky / satellite. For a mobile installation, this means that it is preferable to install at the highest point on the asset where it has a clear view of the sky in all directions.
   - Select an outdoor location where the top of the ST 8100 (the side with the solar panel) has direct exposure to sunlight and the sky.
   - Mount the device so that the device's line-of-sight with the sky is clear of obstructions.
   - Mount the device on the roof of the asset where it has a clear line of sight with the satellite.

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- Mount the device on a flat horizontal surface for mobile installations such that the elevation angle does not change with rotation.
- When deployed on roads (especially trailers), check that the installation meets the maximum vehicle height permitted in the country where the device operates. Maximum vehicle height examples (subject to change and the installer's responsibility to verify these values): Europe: 4 to 6 m, United States of America: 13 ft 6 in. to 14 ft., Canada: 4.15 m, and Kenya and South Africa: 4.3 m.
- Fasten the device securely so that it is not loose and does not move easily.
- Mount the device on a solid, stable surface. If necessary, use a mounting bracket (not supplied) or other suitable support.
- Ensure that any paint above the device is non-metallic and non-metallic flake if the installation is under fiberglass or composite wind fairings.
- Mount the device on the driver's side of the vehicle, if possible, when there is a possibility of strikes by overhanging tree branches.
- Mount the device on a surface that does not get hotter than the maximum operating temperature (70°C/158°F). If the surface may get hotter, mount the device with a thermal barrier between it and the mounting surface.
- DO NOT mount the device on stacked assets / containers.
- DO NOT mount the device close to other electrical equipment due to possible radiated and / or conducted electromagnetic interference.
- DO NOT mount the device on or next to any external sources of heating or cooling.
- DO NOT mount the device close to radar or other communications antennas. Use the following guidelines:
  - > 1 m from VHF/UHF antenna
  - > 3 m from loop antenna
  - > 4 m from MF/HF antenna
  - > 5 m from other satellite antennas
  - Not within a radar beam
- DO NOT mount the device where water may build-up or collect.

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- DO NOT mount the device close to an exhaust pipe due to the excessive heat and the potential for the exhaust pipe causing satellite blockage.
- DO NOT mount the device close to air horns or any tractor roof hardware (for example, emergency lights) that could interfere with satellite communications.
- DO NOT install the device inside a truck under the roof liner.

2. Mark the location of the corners of the ST 8100 on the asset. This is recommended for accurate prepping the asset surface.

1.3 Determine an Installation Method

1. Determine an installation method:
   - Screws or Rivets (permanent mount - recommended method)
   - Magnets and Tether (temporary mount)

1.4 Screws or Rivets Installation

1.4.1 Gather the Required Tools and Materials

Ensure you have the following for this installation:

- ST 8100 Standard Variant (ST8100-0100-H) or ST 8100 Advanced Variant (ST8100-1100-H)
- Mounting hardware (customer provided)
  - For screw installations:
    - 4 mm or #8 size self-drilling screws
    - Drill
  - For rivet installations:
    - 4.8 mm or 3/16" rivets
    - 5 mm (0.196") diameter drill bit
    - Drill
    - Rivet gun
- Clean cloth / towel
- 16-pin cable[1] harness (for advanced variant only, sold separately ST101697-00x)

1.4.2 Prepare the Device (Standard Variant ONLY)

This section only applies if you are installing the ST 8100 Standard Variant (ST8100-0100-H). If installing the advanced variant (ST8100-1100-H) refer to section 1.4.3.

1. If using the tamper magnet, refer to APPENDIX A Prepare the Tamper Magnet before continuing with the installation instructions in section 1.4.4.

[1] This blunt cut cable (ST101697-00x) can be modified for specific applications.

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1.4.3 Prepare the Device (Advanced Variant ONLY)

This section only applies if you are installing the ST 8100 Advanced Variant (ST8100-1100-H). If installing the standard variant (ST8100-0100-H) refer to section 1.4.2.

1. Pull off the cable cover and remove the baggie. The baggie contains a dielectric grease packet and screws. Put these aside for now.



2. Remove the 16-pin black protective cap.
3. Apply dielectric grease to the 16-pin connector.



4. (Advanced Variant only) Connect the 16-pin cable harness to the 16-pin connector. The latches on the cable harness provide a tactile click when engaged. Cable sold separately.

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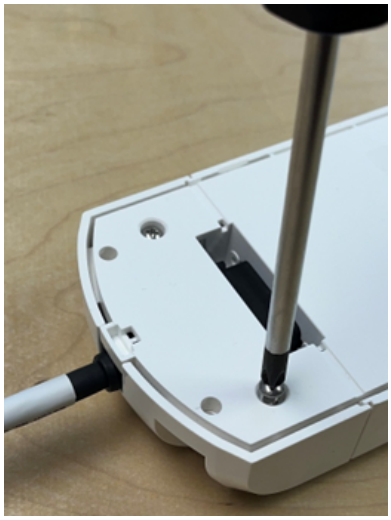


5. Re-attach the cable cover.



6. (Optional) Fasten two (2) screws (included in the baggie) through the underside of the device, into the cable cover.

CAUTION: Screw in by hand, DO NOT over-tighten.



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7. If using the tamper magnet, refer to APPENDIX A Prepare the Tamper Magnet before continuing with the installation instructions in section 1.4.4.

1.4.4 Install the ST 8100 with Screws or Rivets

Mount with Screws

1. Assemble the ST 8100 to the asset with four (4) self-drilling screws (4 mm or #8 size).



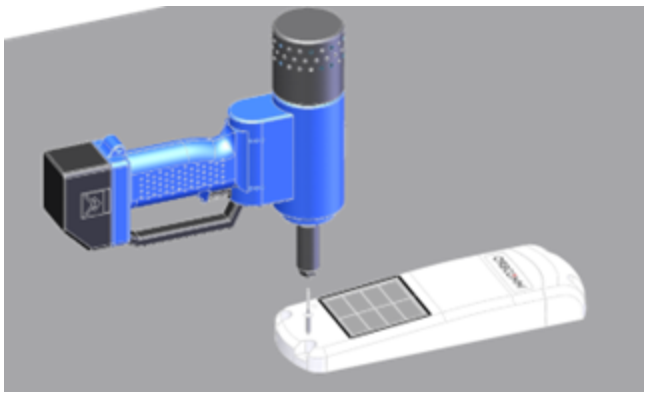
Mount with Rivets

1. Use the ST 8100 as a drill template and drill holes in the asset.

   Note: The drill bit size (4.8 mm or 3/16") is based on the choice of rivet. Use a 5 mm (0.196") diameter drill bit.



2. Use a rivet gun to install four (4) rivets.



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1.5 Magnet Mount Installation

1.5.1 Gather the Required Tools and Materials

Ensure you have the following for this installation:

- ST 8100
- Magnet mount kit with tether (p/n ST100904-002)
- Clean cloth / towel
- Philips screwdriver
- 11/32" socket extension
- 16-pin cable[1] harness (for advanced variant only, sold separately ST101697-00x)

1.5.2 Install the Magnets

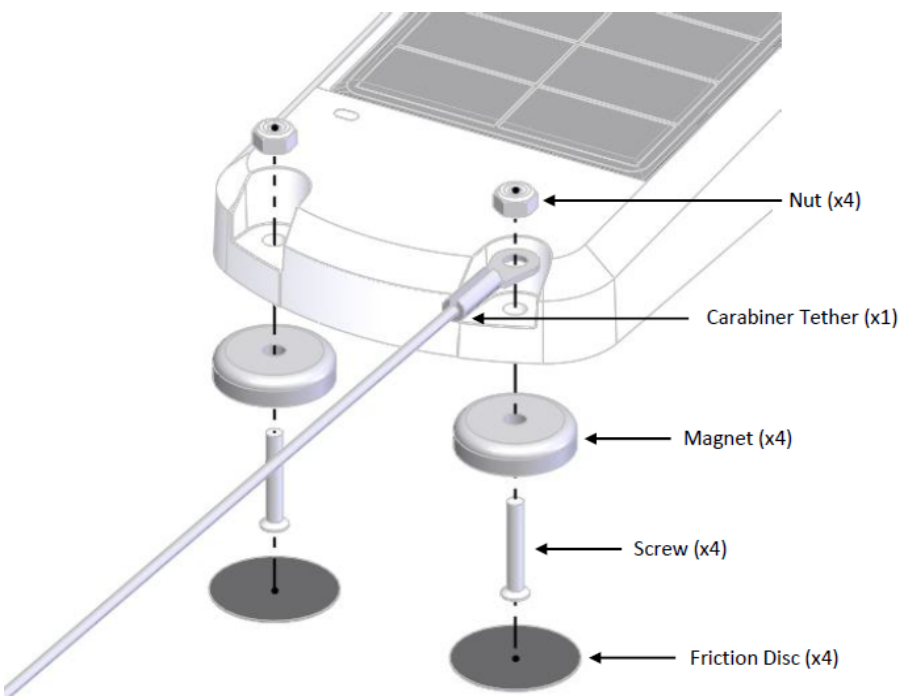
CAUTION: Remember to keep magnets away from electronic devices that store information magnetically such as computers and bank cards.

1. Clean and dry the asset mounting surface before applying the magnet pads.
2. Remove and dispose of the tamper magnet and double-sided tape from the underside of the ST 8100.
3. Assemble the magnets (in four (4) locations) and tether to the ST 8100 as shown below.

   CAUTION: Peel the liner off the friction disc and center it on the magnet. Press down gently so that the magnet pad adheres to the magnet. Ensure that the friction disc completely covers the magnet to reduce the risk of asset damage due to scratches.

[1] This 16-pin cable harness (ST101697-00x) can be modified for specific applications.

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4. Tighten the screws with a hand screwdriver to 1.6 N-m (14 in-lbf).

   CAUTION: DO NOT over-tighten the screws.

5. Attach the tether to the ST 8100 using one of the magnet mount screws.



Note: When using the tether, assemble the eyelet of the tether through one of the screws holding the ST 8100 in place.

6. Place the ST 8100 on the asset where it has a clear line-of-sight to the sky and then take the carabiner end of the tether and either wrap it around a secure location / post and clip it back to the tether cable (figure A) or clip it directly to a secure fixture (figure B).

Note: Figures are for illustration purposes only.

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(A) Clip back to tether



(B) Clip to a secure fixture



APPENDIX A PREPARE THE TAMPER MAGNET

A.1 Gather the Required Tools and Materials

Ensure you have the following for this installation:

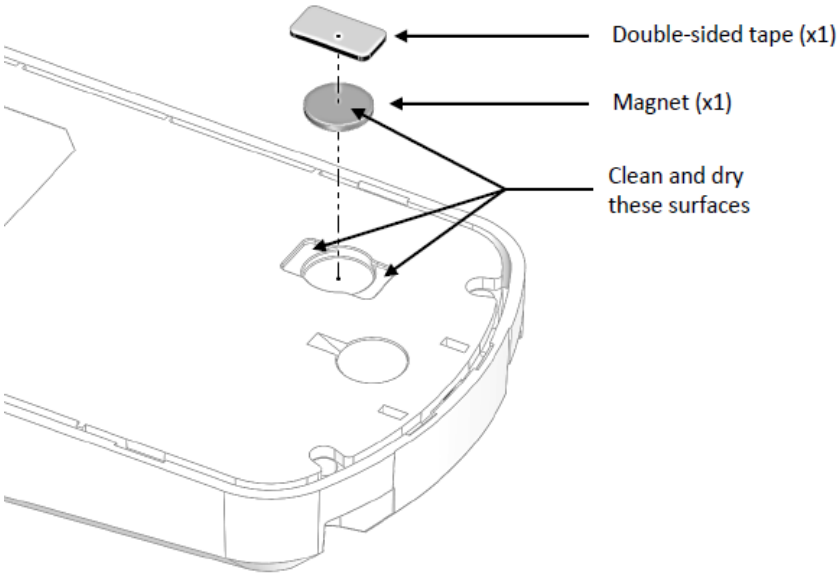
- Tamper Magnet Kit (p/n ST101833-001)
- Clean cloth / towel
- Isopropyl alcohol or an equivalent product

A.2 Review the Installation Guidelines

- The tamper magnet is the last step before fastening screws / rivets.
- For rivet installations, use the ST 8100 as a template and drill the holes to prepare for the installation of rivets.
- The asset must be clean and dry where the magnet bonds with the asset.

A.3 Install the Tamper Magnet

1. Clean the asset surface with isopropyl alcohol or an equivalent product that does not leave a residue and wait for the surface to dry completely.
2. Clean and dry the magnet and the magnet recess on the device.



3. Insert the magnet into the recess.
4. Remove one of the liners from the double-sided tape and press it over the magnet in the recess for at least 10 seconds to bond the magnet to the device.

   CAUTION: DO NOT touch the adhesive.

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5. Remove the second liner before pressing firmly on the surface of the device, for at least 10 seconds, to bond the device to the asset.

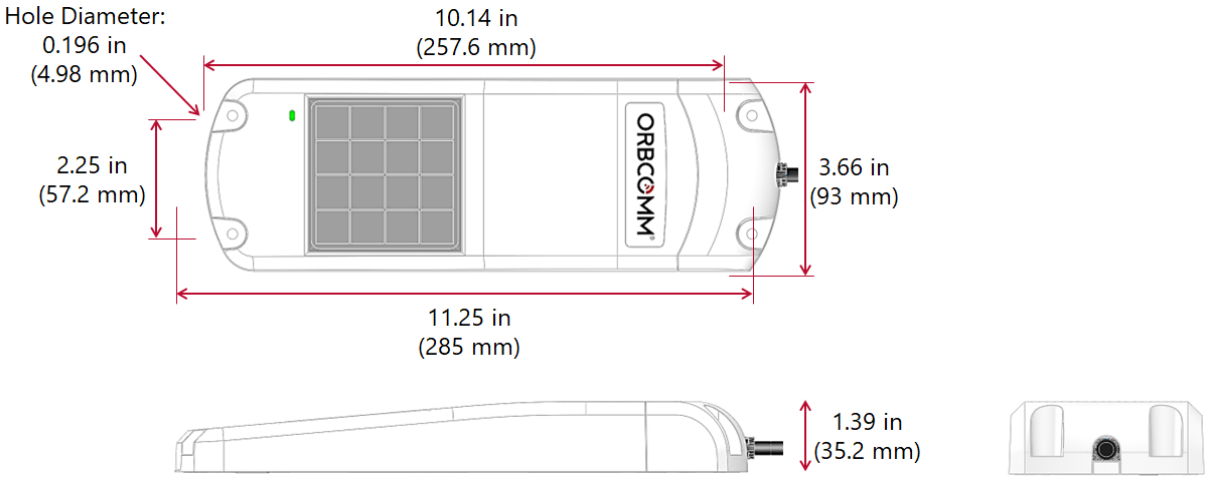
   Note: For rivet installations, use a rivet to align the ST 8100 with the previously drilled rivet holes.

6. Hold the device in position to prevent it from moving while you fasten the mounting hardware.

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APPENDIX B MECHANICAL DIMENSIONS

The ST 8100 mechanical dimensions are shown in inches and (mm).



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APPENDIX C 16-PIN CABLE HARNESS

The advanced variant has a 16-pin connector. The connector allows various peripherals such as RS-232/RS-485, configurable I/O, high voltage analog inputs, and analog current inputs. The cable harness to interface with the connector is part number: ST101697-00X.

Table 1 maps to the layout shown in Figure 1.

Figure 1: View of Connector

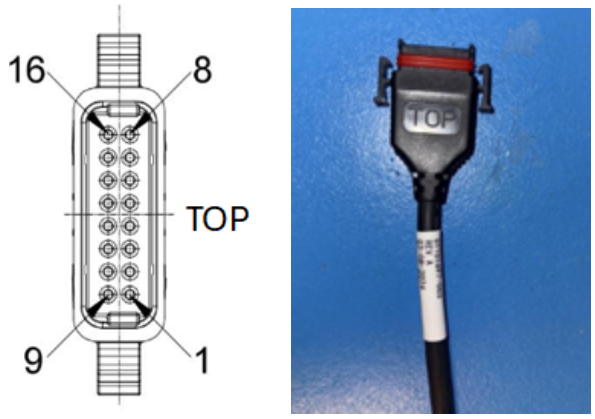


Table 1: Electrical Pin Assignment

PIN	Function	Wire Color
1	I/O_2 OUT	Yellow
2	Ground	Short circuit with pin 10
3	24 V OUT 1	Black
4	Current ADC Input 1 (I IN 0)	Brown
5	RS-485_B	Blue
6	RS-232 TX	Gray
7	Voltage ADC Input 1 (HV Analog 1)	White with a black trace
8	1-Wire I/O	Red with a black trace
9	I/O_1 OUT	Green
10	Ground	Short circuit with pin 2
11	24 V OUT 2	Red
12	Current ADC Input 2 (I IN 1)	Orange
13	RS-485 A	Purple
14	RS-232 RX	White
15	Voltage ADC Input 2 (HV Analog 2)	Brown with a black trace
16	Ground	Orange with a black trace

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